

Clockless Federated Adaptation

無全域時鐘下的非同步醫學影像聯邦學習

Group 4 : Asynchronous Distributed ML Adaptation

分散式系統期末 Proposal Deck

02 01 Motivation | 從醫院合作訓練開始

多家醫院想共同訓練醫學影像模型，但原始資料不能集中。

- 醫院設備、網路、工作負載不同。
- 有些 update 很快，有些 update 很晚。
- 病人族群與影像分布不同。

問題：不等慢醫院會比較快，但晚到的 update 還可信嗎？

03 01 Motivation | 分散式系統核心矛盾

Sync FL: 等所有 client -> 最慢節點決定速度

Async FL: 不等慢 client -> 收到 stale / out-of-order update

本專案聚焦：

throughput vs consistency / convergence correctness

這是 distributed systems 的時間、順序與一致性問題，不只是分類模型問題。

04 02 Preliminary | Sync vs Async FL

Setting	System behavior	Main risk
Sync FedAvg	round barrier , 等待 clients	straggler bottleneck
Naive Async	update 到就套用	stale update instability
Buffered Async	收 B 個 updates 再聚合	buffer 中仍可能有 conflict

05 02 Preliminary | Logical Staleness

不假設 synchronized physical clock。

```
server_version = current global model version  
client_start_version = version received by the client  
staleness = server_version - client_start_version
```

這是 Lamport logical time 在 FL simulator 裡的對應：用 logical version 取代 wall-clock ordering。

06 03 Problem Formulation | 公平比較

Async 以前表現差，可能只是 update budget 少。

```
Sync update budget  = rounds × clients  
Async update budget = events  
Fair comparison     = events = rounds × clients
```

正式矩陣：

```
9 datasets × 6 methods × 3 seeds = 162 official runs
```

07 03 Problem Formulation | 評估指標

ML 指標：

```
best_acc, final_acc, stability_drop = best_acc - final_acc
```

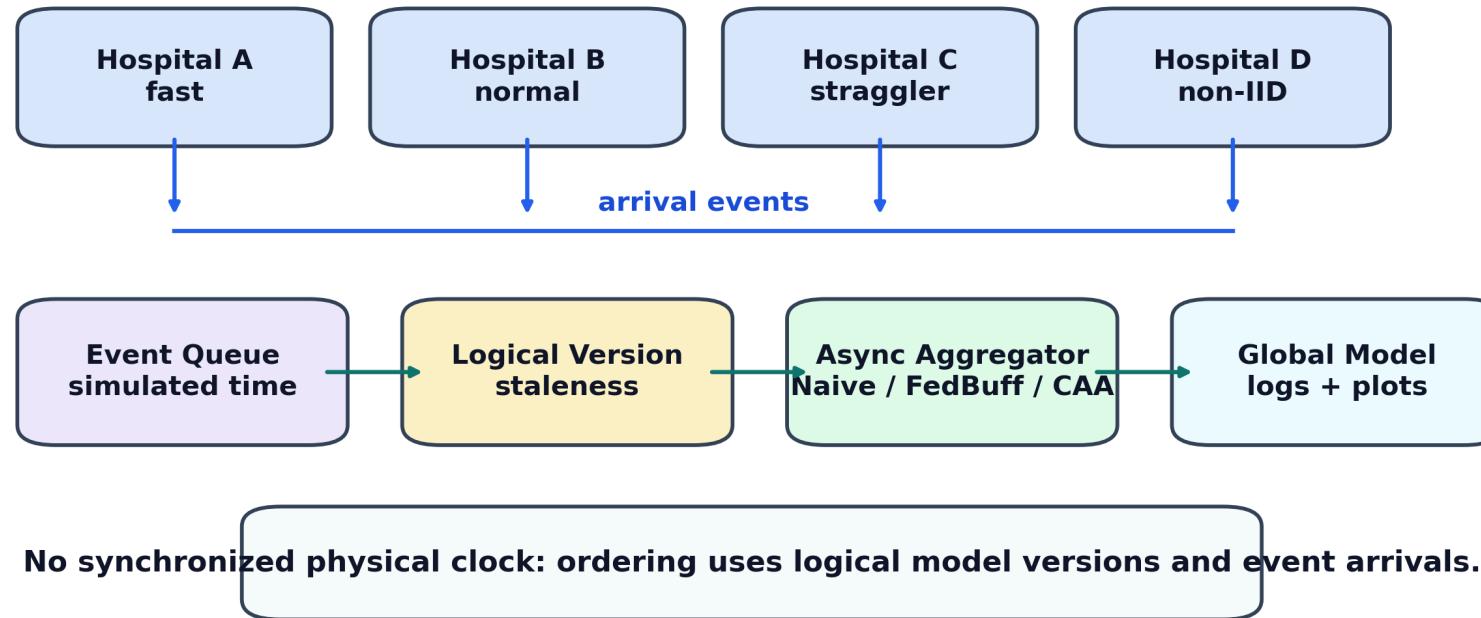
Distributed systems 指標：

```
p95 staleness, simulated time, effective alpha,  
client contribution Gini, time-to-accuracy
```

accuracy 是結果；staleness / delay / imbalance 解釋原因。

08 03 Architecture | Event-Driven Clockless Simulator

Event-Driven Clockless FL Simulator

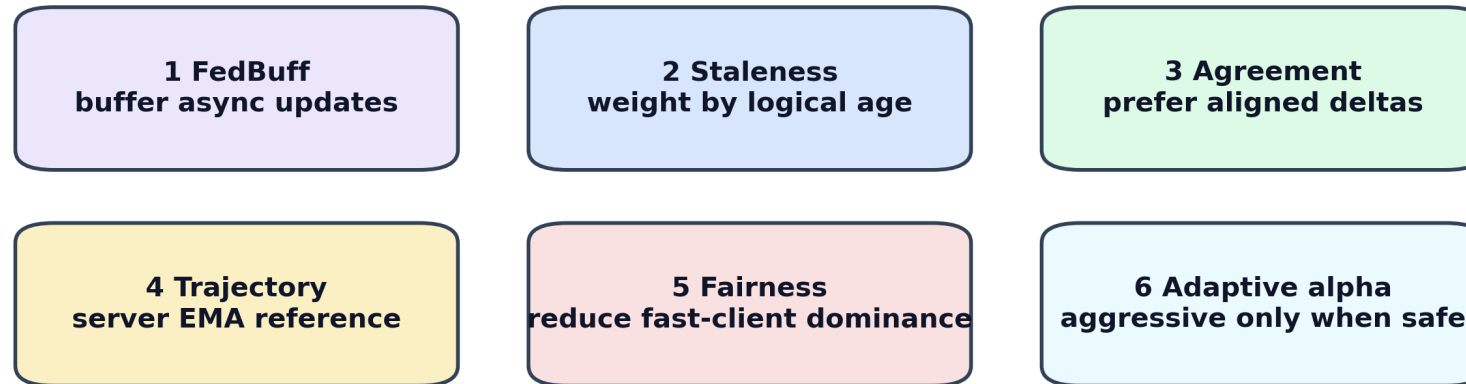


09 04 Methodology | Baseline Map

Method	Role	What it tests
Sync	upper reference	no async staleness, but barrier
Naive Async	stateless async	throughput without correction
Staleness	logical-age correction	whether age alone is enough
FedBuff	buffered async baseline	whether buffering stabilizes
CAA-v1	agreement-aware	whether direction helps
CAA-v2	final method	direction + trajectory + fairness

10 04 Methodology | CAA-v2 不只是 FedBuff

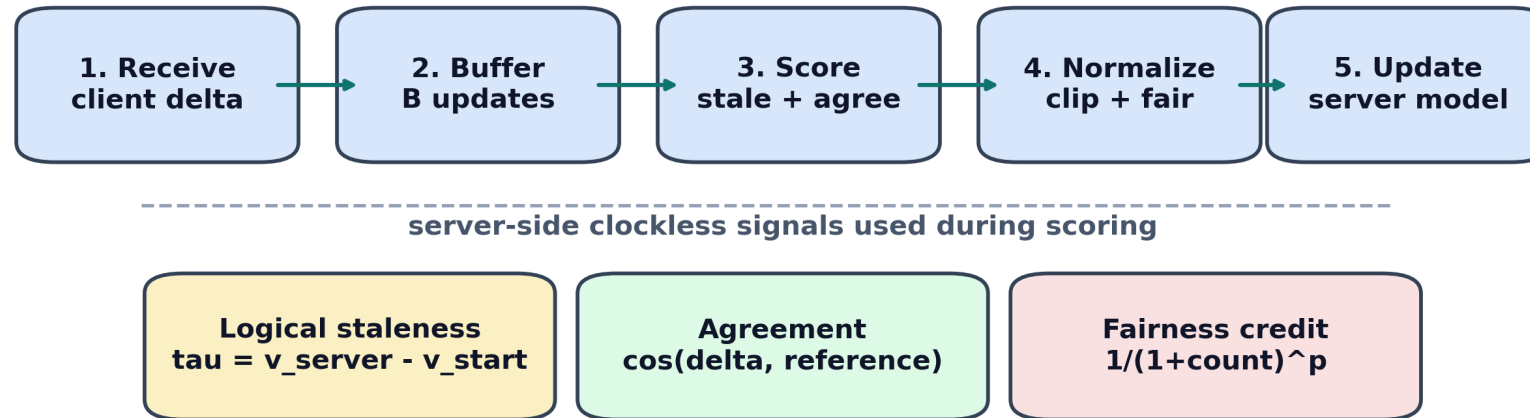
What CAA-v2 Adds Over FedBuff



Simple server rule: combine known signals into one clockless async aggregation policy.

11 04 Methodology | CAA-v2 Server Pipeline

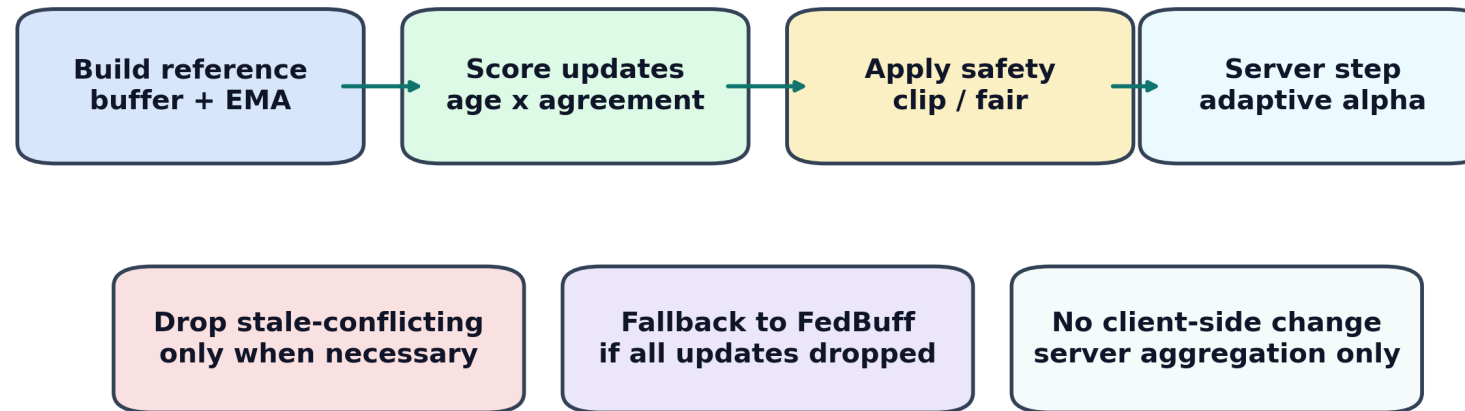
CAA-v2 Aggregation Pipeline



All decisions use logical versions, deltas, accepted trajectory, and contribution counts.

12 04 Methodology | Server Decision Flow

CAA-v2 Server Decision Flow



The flow is deterministic and clockless; no physical timestamp is required.

13 04 Methodology | 核心數學式

每個 client update :

```
delta_i = client_model_i - model_at_client_start_i  
tau_i   = server_version - client_start_version_i
```

CAA-v2 權重 :

```
raw_weight_i = n_i × staleness_decay(tau_i)  
              × agreement(delta_i, ref) × fairness_i
```

Server update :

```
w <- w + alpha_buffer × weighted_average(delta_i)
```

14 04 Methodology | Reference Direction

```
buffer_ref = weighted_average(delta_i, n_i × age_i)
server_ema = EMA(previous accepted server deltas)
ref        = blend(buffer_ref, server_ema)
```

直覺：

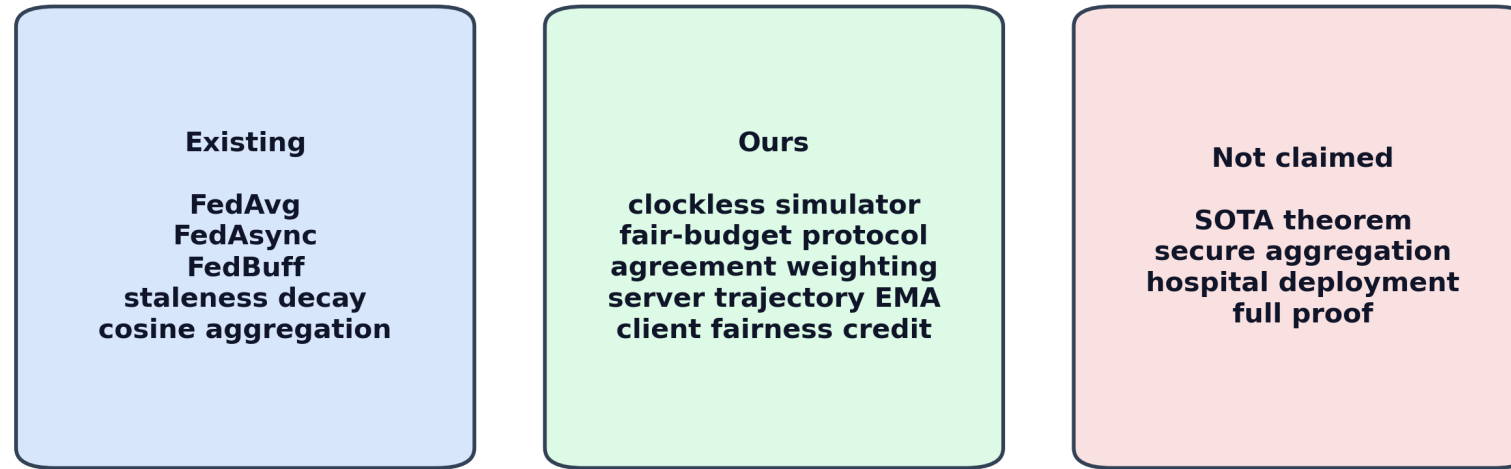
- buffer_ref：這批 updates 的群體方向。
- server_ema：近期 global model 真的往哪裡走。
- stale 且反向的 update 不應被盲目放大。

15 04 Methodology | Clockless Signals

Signal	Source	Uses physical clock?
staleness	server/client logical version	No
agreement	delta dot product	No
server trajectory	accepted server deltas	No
fairness credit	accepted contribution count	No
adaptive alpha	mean staleness + agreement	No

16 04 Methodology | Novelty Boundary

Novelty Boundary



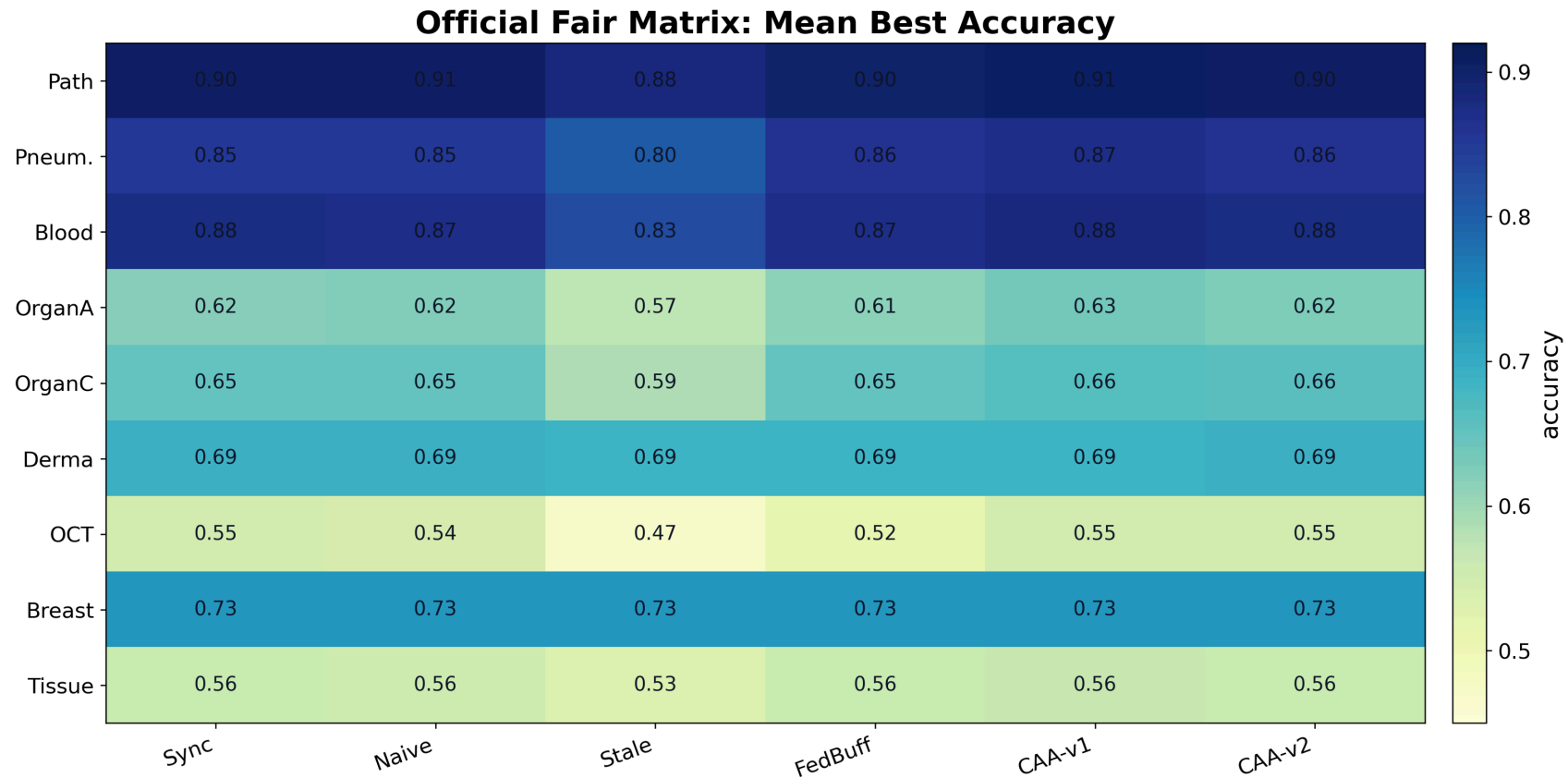
Contribution: reproducible systems integration and evidence, not a wholly new FL primitive.

17 05 Experiment Results | Fairness Protocol

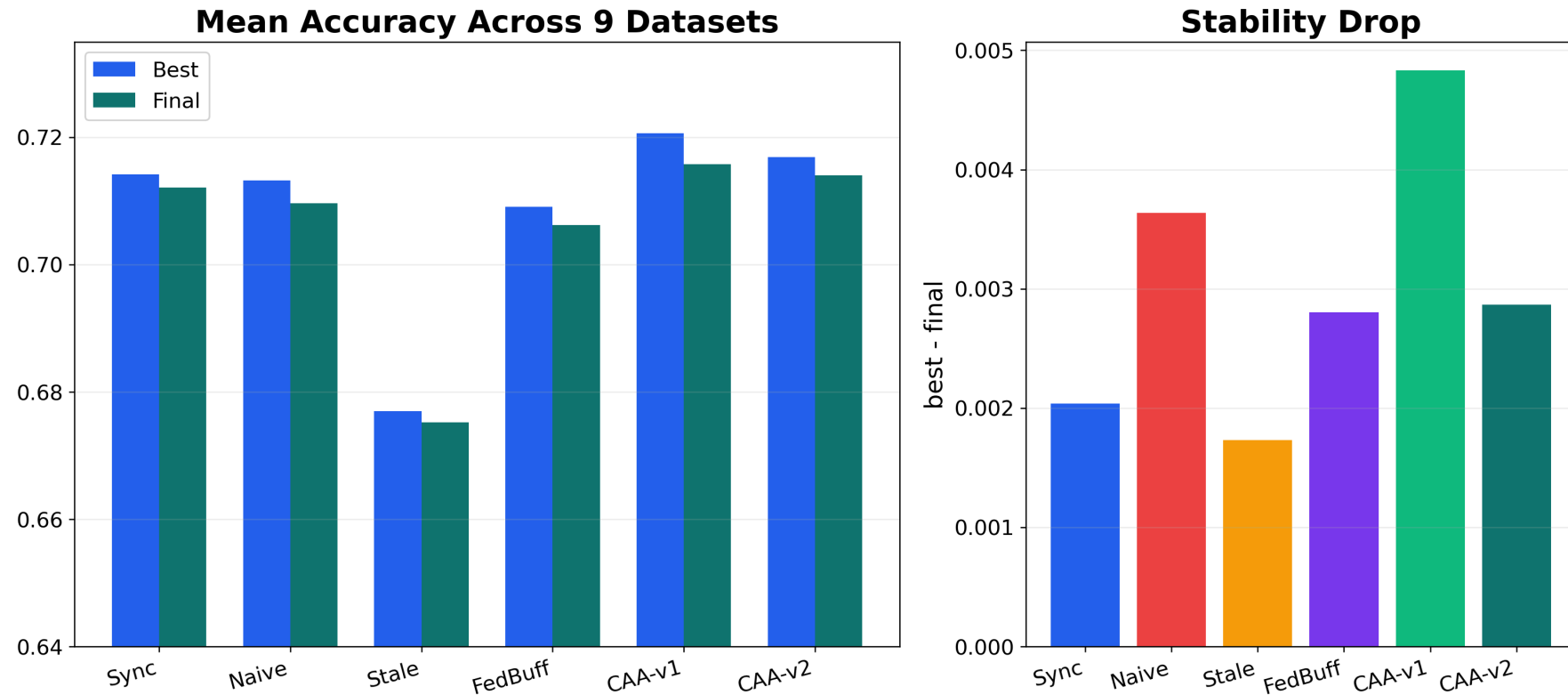
```
clients = 10
local_epochs = 1
batch_size = 128
lr = 0.01, cosine scheduler
seeds = 42, 43, 44
async delay = same heterogeneous setting
fair budget = async events = sync rounds × clients
```

正式比較不混用不同 backbone、delay、seed、local epochs 或 update budget。

18 05 Experiment Results | Coverage

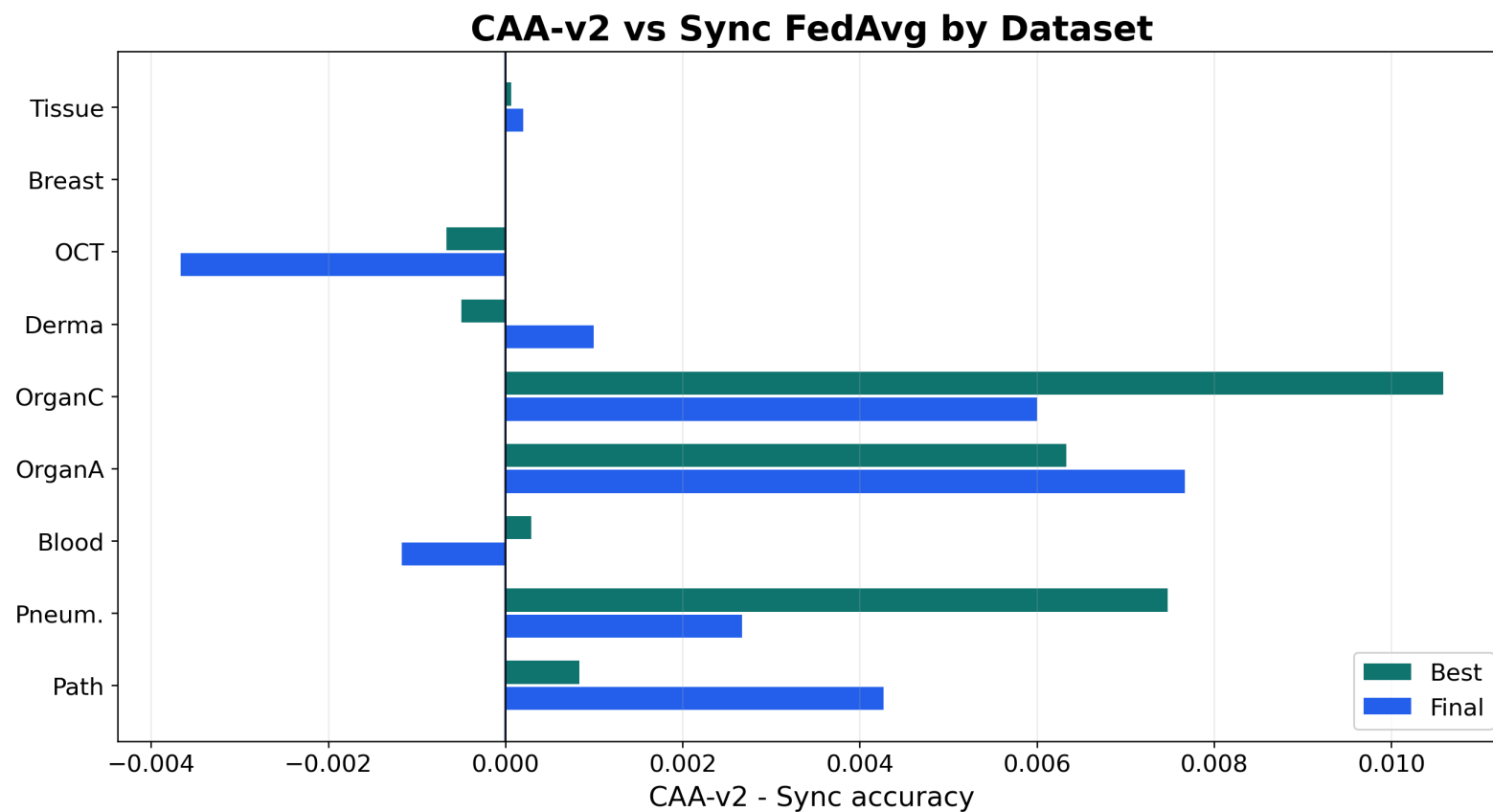


19 05 Experiment Results | Overall Dashboard

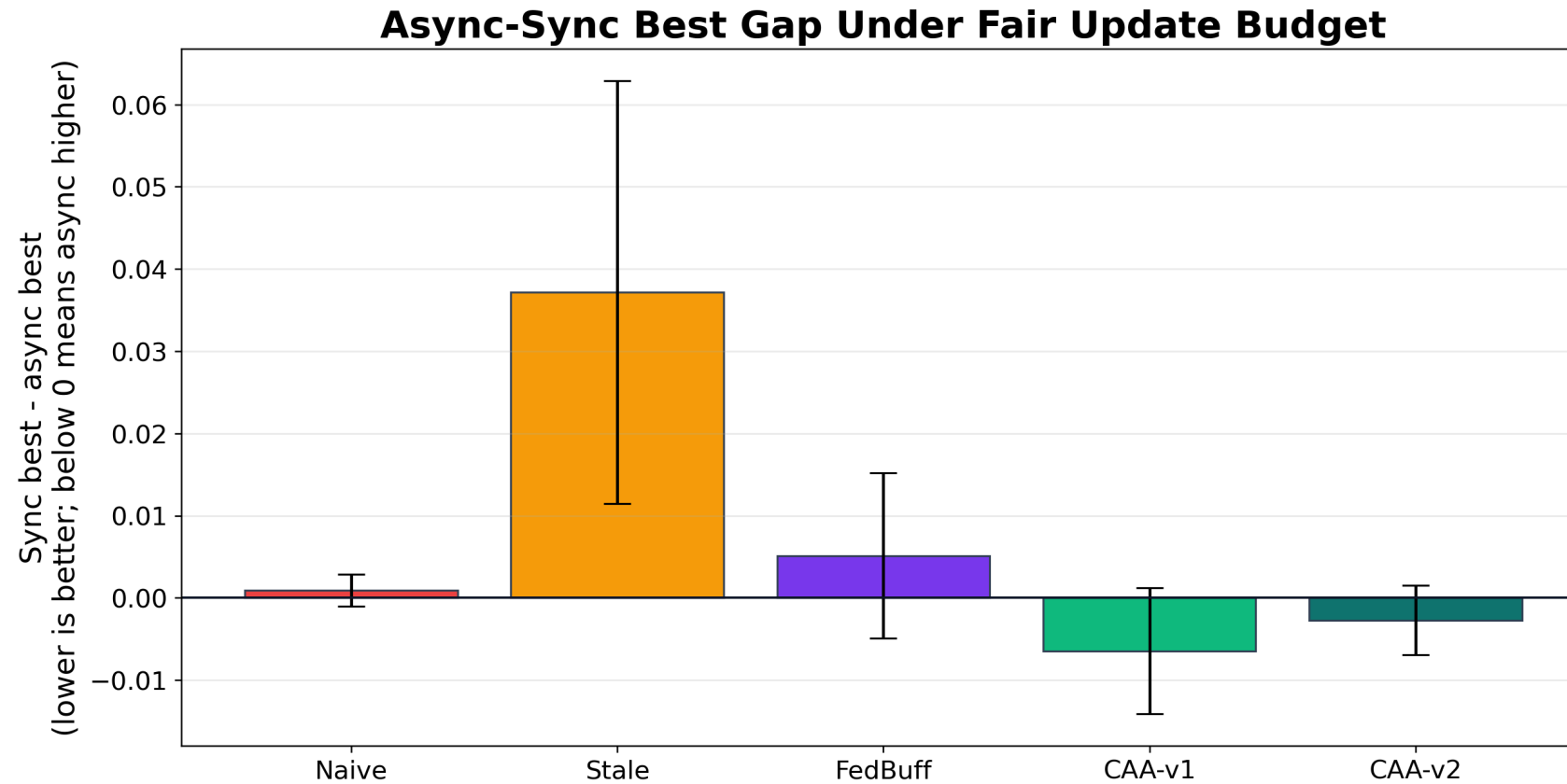


Takeaway: CAA-v2 improves mean final accuracy over Sync/Naive while keeping drop below CAA-v1 and Naive.

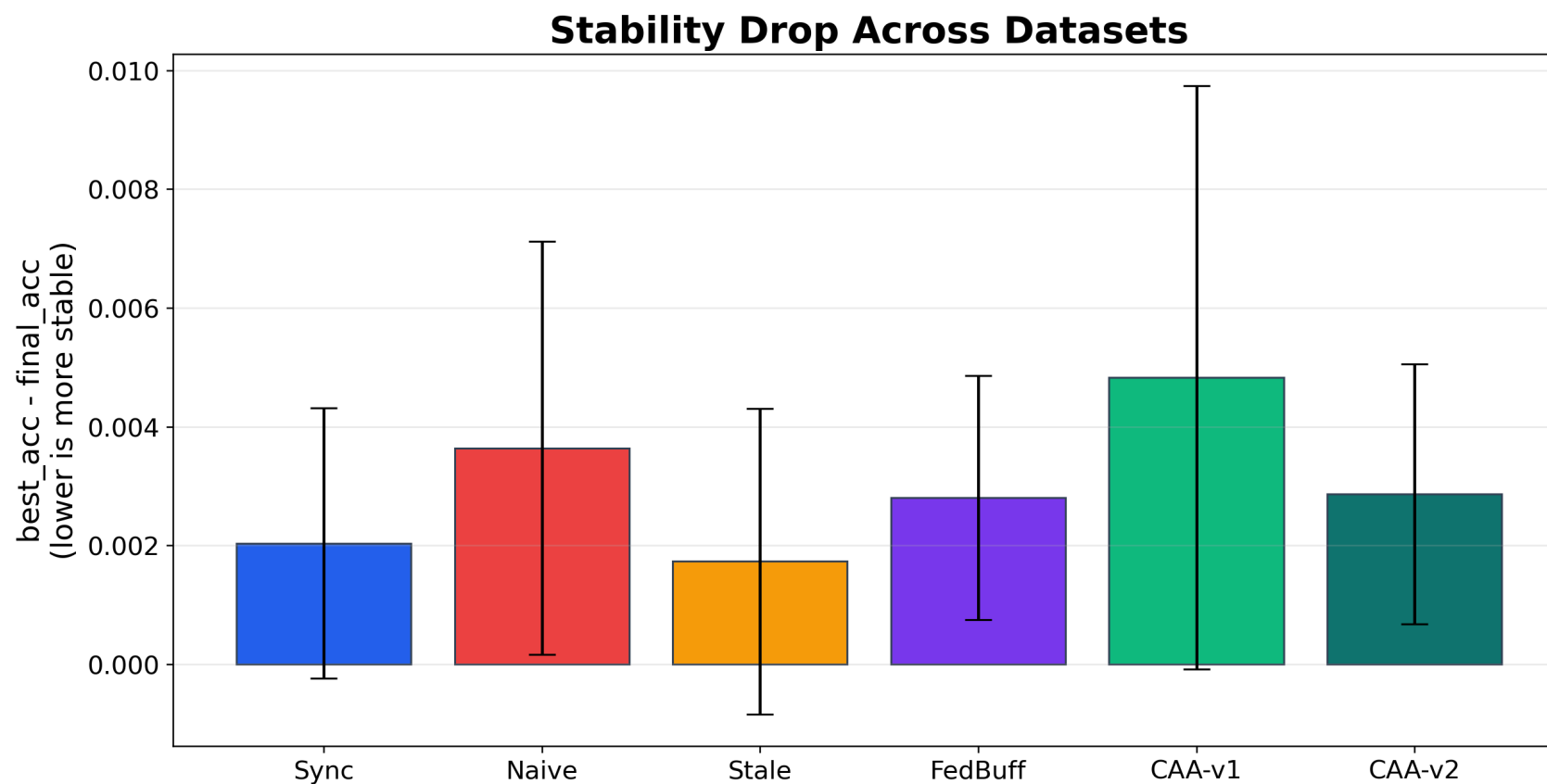
20 05 Experiment Results | CAA-v2 vs Sync



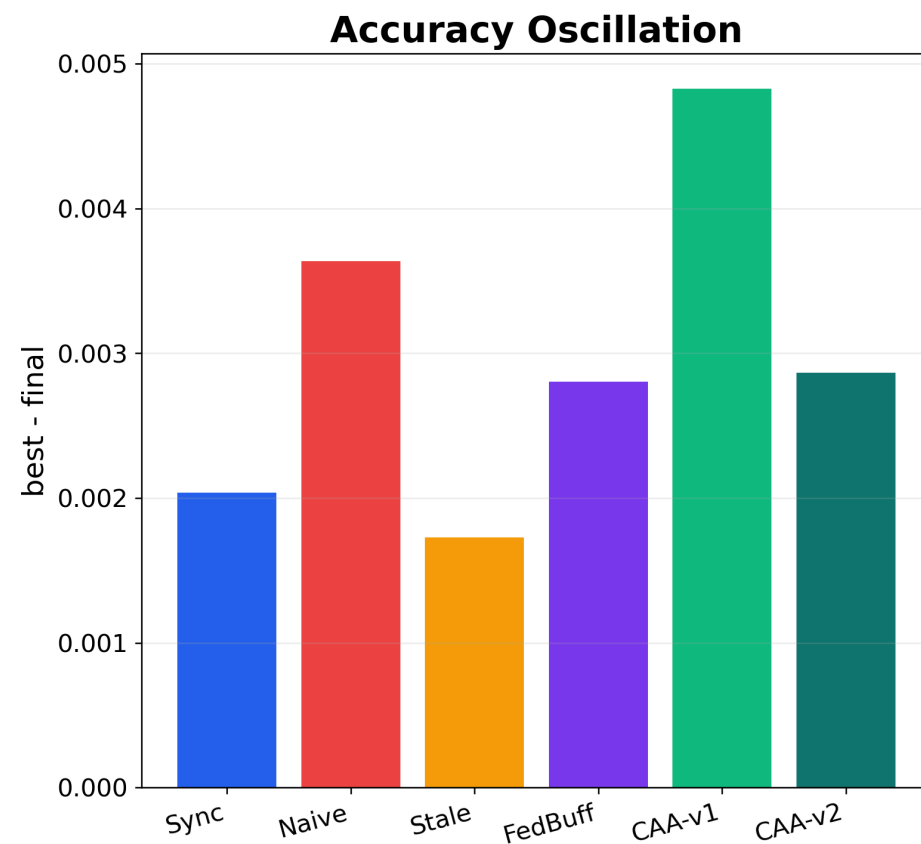
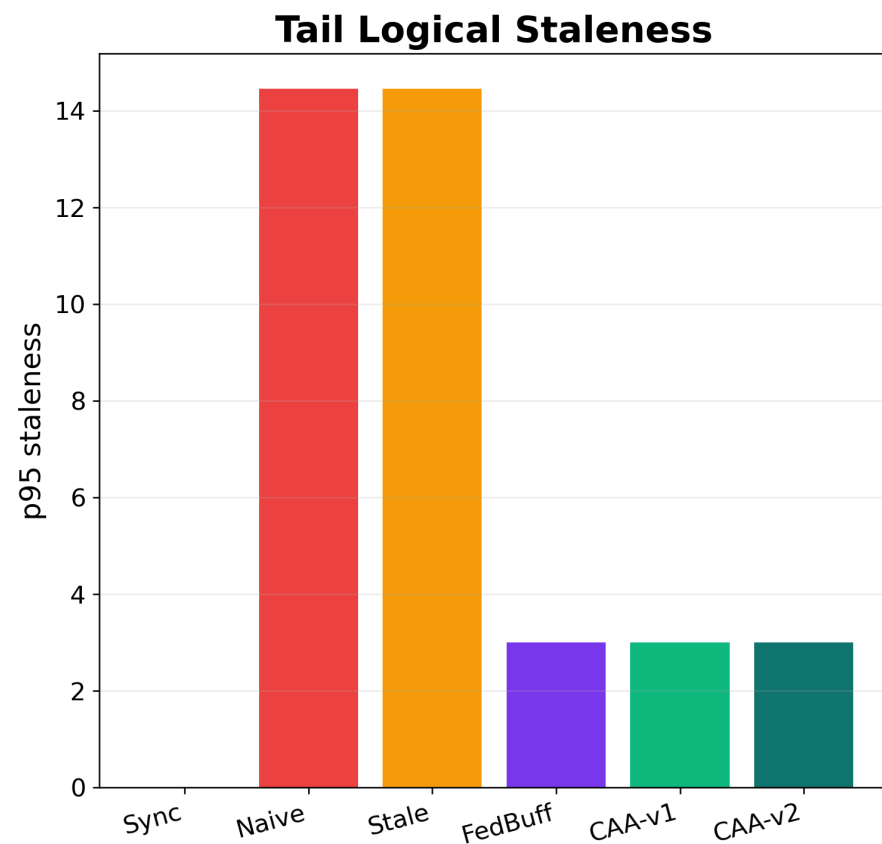
21 05 Experiment Results | Async-Sync Gap



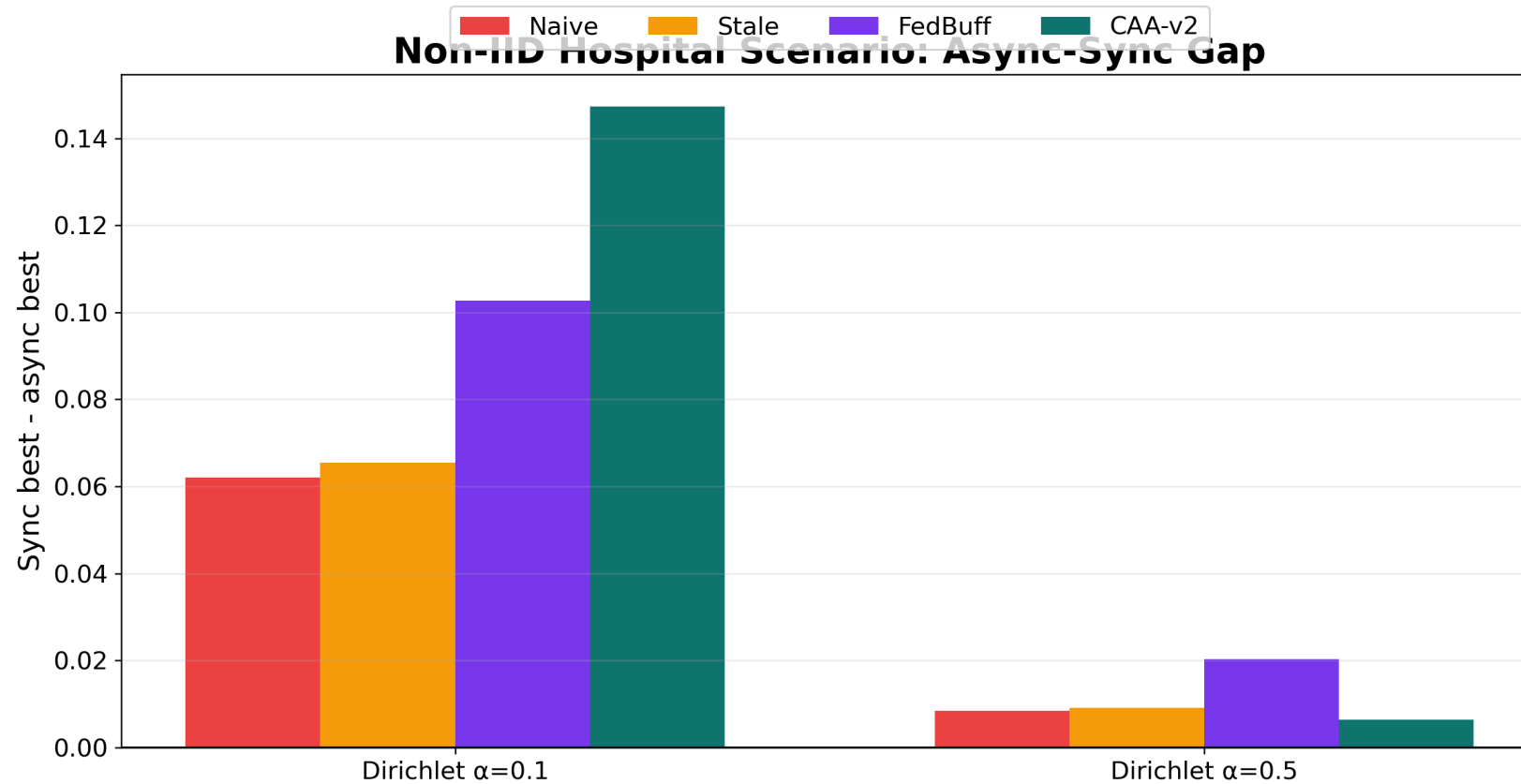
22 05 Experiment Results | Stability Drop



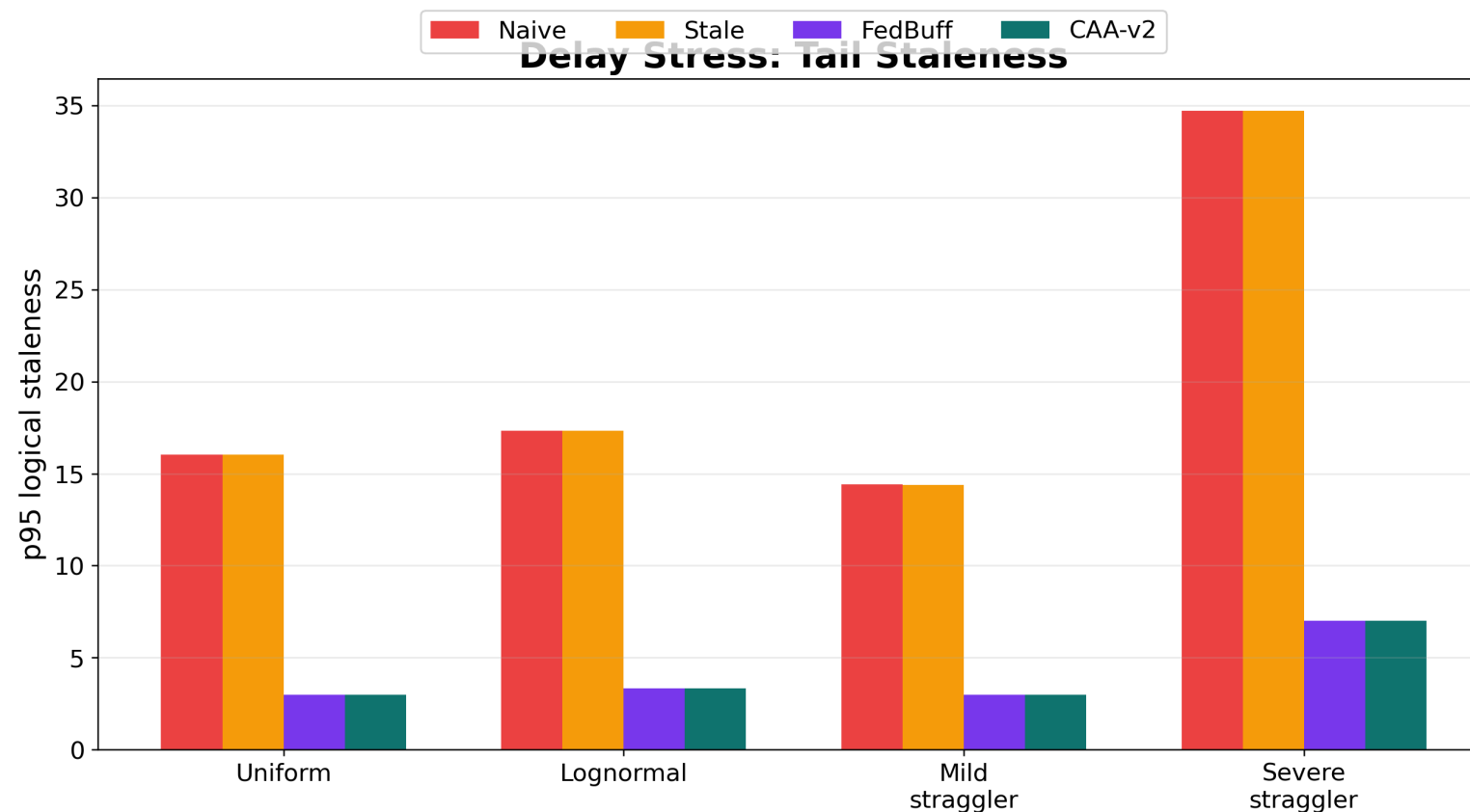
23 05 Experiment Results | System Metrics



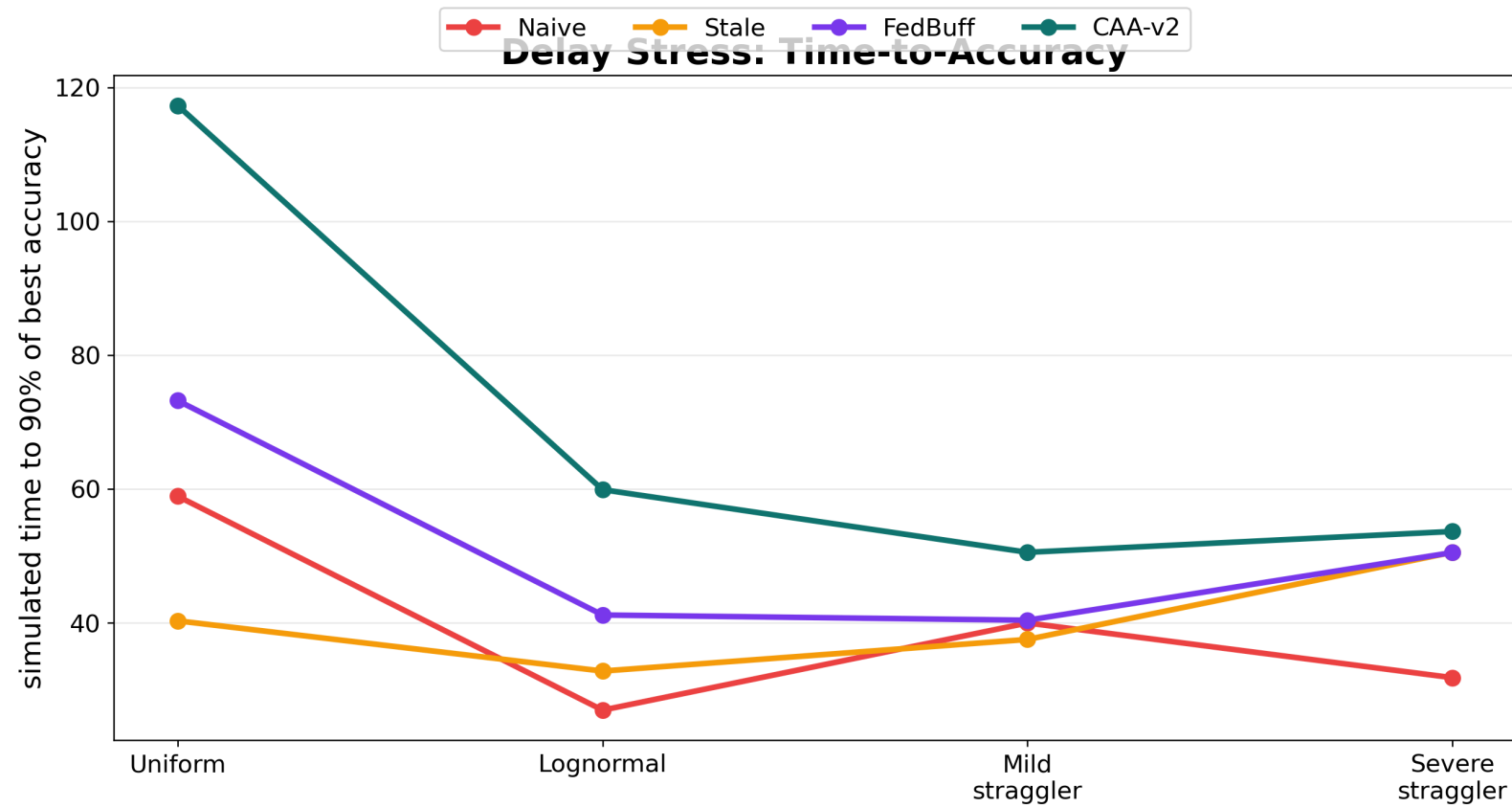
24 05 Experiment Results | Non-IID Hospital Scenario



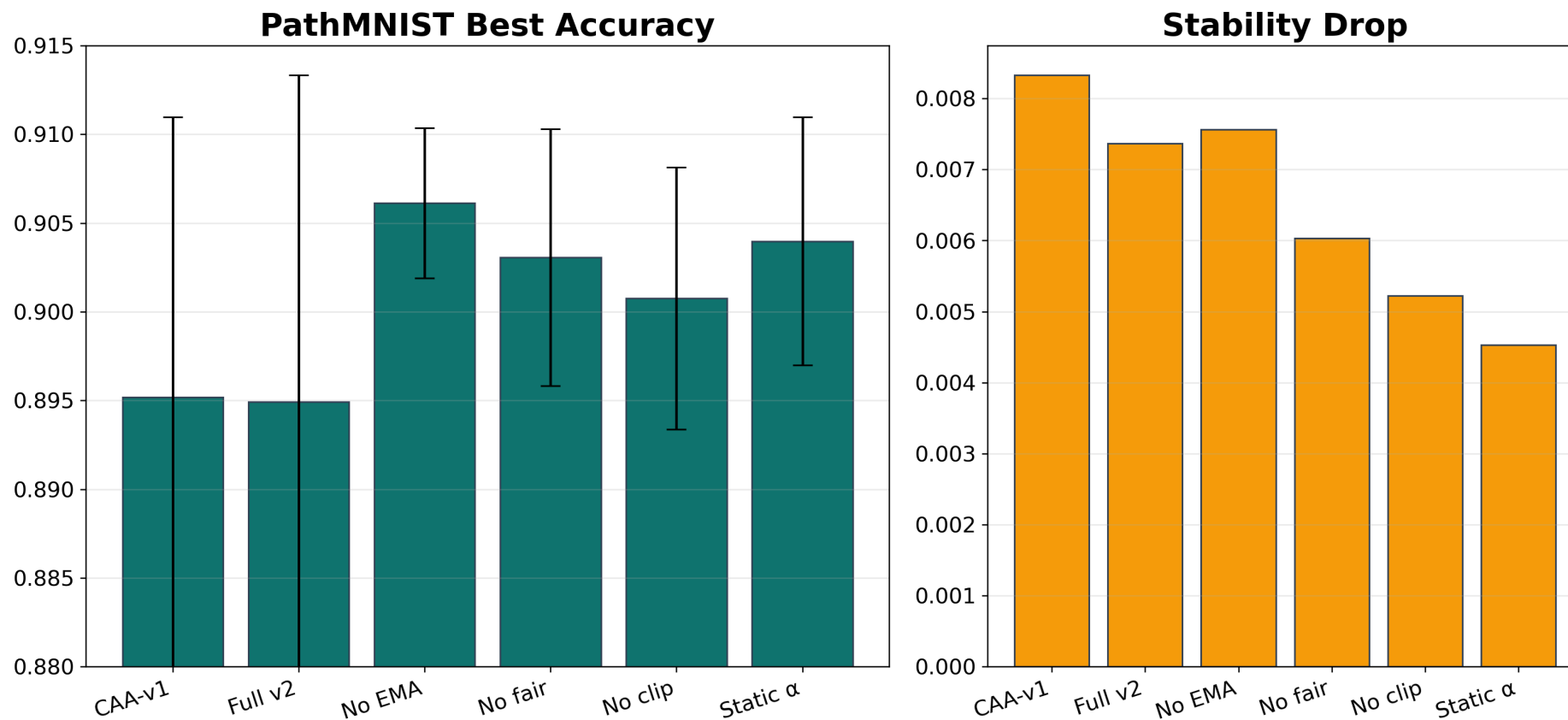
25 05 Experiment Results | Straggler Stress



26 05 Experiment Results | Simulated Time



27 05 Experiment Results | Ablation



28 05 Experiment Results | Final Interpretation

正式 fair matrix :

```
CAA-v2 best  = 0.7169  
CAA-v2 final = 0.7140  
Sync final   = 0.7121  
Naive final  = 0.7096
```

結論：CAA-v2 不是 universal winner，但它比 Naive 穩、比 staleness-only 不保守，並且在多數資料集接近或超過 Sync。

29 06 Challenges | 總覽

Distributed-Systems Challenges Made Visible

No global clock
no total physical order

Stale updates
old model version

Conflicting updates
non-IID hospital data

Fast-client bias
arrival dominance

Report accuracy + staleness + simulated time + client imbalance

30 06 Challenges | No Global Clock, Stale, Conflict

三個問題會同時發生：

```
No global clock -> cannot totally order distributed events  
Stale update    -> trained from old global model  
Conflict update -> non-IID direction differs from others
```

CAA-v2 的設計：logical staleness + direction agreement + server trajectory。

31 06 Challenges | Fast Clients 與 Privacy Tension

Fast-client domination :

快醫院 arrival rate 高 -> 可能長期主導 global model

Privacy tension :

CAA-v2 需要 delta direction / norm
secure aggregation 相容性仍需未來設計

所以目前主張是 system simulator / design extension , 不宣稱完整 privacy solution 。

32 07 Conclusion | 貢獻總結

1. 用 logical version 建立 no-global-clock async FL simulator。
2. 補上 fair update-budget protocol。
3. 提出 CAA-v2：agreement + server trajectory + fairness credit。
4. 用 9 datasets $\times$ 6 methods $\times$ 3 seeds 驗證。
5. 用 staleness、simulated time、client Gini 補足 distributed systems 分析。

33 07 Conclusion | Final Claim

Under a fair update budget, CAA-v2 makes clockless asynchronous FL approach Sync FedAvg across diverse MedMNIST datasets, while reducing the instability of Naive Async and avoiding the over-conservatism of staleness-only aggregation.

中文：

CAA-v2 讓無全域時鐘的非同步 FL 接近 Sync FedAvg，同時比 Naive Async 穩、比 staleness-only 不保守。

34 07 Future Work

- 對比 FedStaleWeight、SEAFL、FedPSA、FedCompass。
- non-IID / straggler stress 增加 multi-seed。
- 加入 real network / hospital-like traces。
- 補 balanced accuracy、macro-F1、minority recall、AUROC。
- 設計 secure-aggregation-compatible agreement statistics。
- 加入簡化 convergence / stability analysis。

35 References

- Lamport, *Time, Clocks, and the Ordering of Events in a Distributed System*, CACM 1978.
- McMahan et al., *Communication-Efficient Learning of Deep Networks from Decentralized Data*, AISTATS 2017.
- Xie et al., *Asynchronous Federated Optimization*, 2019.
- Nguyen et al., *Federated Learning with Buffered Asynchronous Aggregation*, 2021.
- Ma et al., *FedStaleWeight*, 2024.
- Yang et al., *MedMNIST v2*, Scientific Data 2023.